Iglesia lecture 2

Thursday, 5 October 2023 10:35 AM

NMR → tells you where in the molecule the isotope is.

Early on → used microwave rotational spectroscopy (C-D rotation different from C-H)

Gringer : chemistry to make ^{13}C labels on carbon chains.

Fisher-Tropsch synthesis

Chain initiation ; $CO + H_2 \rightarrow$ olefins + alcohols

↳ coming up w/ C_1 addition (one by one addition)

$R \xrightarrow{k_1} B^* \xrightarrow{k_2} P$ (Radioactivity was independent of chainsize)

^{14}C is starting off & then carbons being added

rate constant ↑ intermediate ↑

SS rate: $r_{ss} = k_2 [B^*]$ } unable to distinguish

↳ transient experiments can decouple the k & $[B^*]$

↳ $\frac{d[P]}{dt} = k_2 [B^*]_{ss} \exp\{-k_2 t\}$

get this from decay

+ The "seed model" of Fisher-Tropsch

Desorption / Adsorption Issue

↳ How do you stop molecules re-adsorbing?

Have isotopic gas flowing → displacing the ^{12}C on surface w/ ^{13}C

Oxidative Dehydrogenation of Propane

Homolytic C-H activation }

$$r = \frac{A}{[C \cdot D]}$$

TS of RDS step

C & D ↓ ↓ averages

RDS step is a step of point of no return

↓

(while the other elementary steps will be at equilibrium)